

# KAVYA PATEL

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## OBJECTIVE

Final-year B.Tech IT student with hands-on experience building and deploying **multi-agent AI systems**, **LLM-powered applications**, and **REST APIs** to production. Shipped projects spanning **LangGraph agent orchestration**, **RAG pipelines**, computer vision, and ML-based recommendation systems, with end-to-end ownership from architecture through **CI/CD deployment**. Skilled at debugging production issues using observability tools and translating research-grade AI techniques into reliable, user-facing products. Seeking an **AI Engineer Internship** to build impactful systems from day one and grow alongside a team solving real-world problems with AI.

## SKILLS SUMMARY

|                                    |                                                                                 |
|------------------------------------|---------------------------------------------------------------------------------|
| <b>AI &amp; LLMs:</b>              | LangChain, RAG, AI Agents, Prompt Engineering, Groq API, OpenAI API, Gemini API |
| <b>Machine Learning:</b>           | Scikit-Learn, PyTorch, Pandas, NumPy, Data Preprocessing, Model Evaluation      |
| <b>Backend &amp; APIs:</b>         | FastAPI, REST APIs, Python, Pydantic, JWT Authentication                        |
| <b>Databases &amp; Vector DBs:</b> | PostgreSQL, MySQL, Supabase, Qdrant, FAISS                                      |
| <b>Frontend:</b>                   | React, Tailwind CSS, Streamlit, HTML, CSS                                       |
| <b>DevOps &amp; Tools:</b>         | Docker, Git, GitHub Actions, LangSmith, VS Code                                 |

## WORK EXPERIENCE

### AI/ML Intern | BISAG-N | Gandhinagar

January 2026 – April 2026

- Developed a dual-approach **Change Detection system** analyzing land-use shifts across Ahmedabad (2017–2025) using 9 Sentinel-2 satellite images (~10.7 GB)
- Engineered an AI pipeline with **SegFormer architecture**, achieving **73% accuracy** across 6 land-use classes (urban, vegetation, water gain/loss)
- Led GIS-based Approach 2** independently within a 5-member team using NDVI/NDBI/NDWI indices, Otsu thresholding, and custom 60% polygon overlap logic
- Deployed end-to-end web platform with **FastAPI + MongoDB** for real-time inference; published spatial maps via GeoServer and Felt.com for QGIS integration

## PROJECTS

### Multi-Agent Research Assistant | [Live Demo](#) | *React, FastAPI, LangGraph, Groq, Tavily, BeautifulSoup, Render, Vercel*

- Architected and deployed a production-grade **7-node multi-agent research pipeline** using **LangGraph StateGraph**, **FastAPI**, and **Groq LLaMA-3.3-70B**, automating end-to-end research (Planner → Search → Reader → Writer → Critic → Fact-Checker → Visualizer) with live web search via **Tavily API** and parallel scraping using **aiohhttp + BeautifulSoup**.
- Strengthened application security by implementing **15+ prompt injection guardrails**, **Pydantic v2** input validation, **SlowAPI** rate limiting, and environment-based secret isolation, ensuring secure API interactions with zero sensitive data exposure in logs or responses.
- Developed a responsive **React + Tailwind CSS** frontend with source-verified visualizations using **Recharts**, one-click searchable PDF export via **@react-pdf/renderer**, and integrated **LangSmith** for end-to-end tracing, debugging, latency monitoring, and agent execution observability.
- Implemented production-ready DevOps with **GitHub Actions**, **pytest (61 automated tests)** with mocked LLM/API calls, and **Vitest**, enabling automated CI/CD, dependency auditing, and build validation for reliable deployments.

### Enterprise Document Intelligence Dashboard | [Github](#) | *Python, Celery, Redis, Qdrant, Streamlit, Docker, Supabase, Groq*

- Built a **production-grade RAG platform** enabling enterprises to query **100+ page** business PDFs conversationally, returning **page-cited** answers
- Eliminated UI freeze on documents up to **100MB** by offloading PDF parsing to a **Celery + Redis** async pipeline, fully decoupled from the main thread
- Achieved high-accuracy retrieval across **multi-page corpora** by combining **semantic vector search** (Qdrant) with keyword matching and **ML reranking** via Groq LLaMA3-70B
- Isolated document libraries for **10+ concurrent users** via Supabase Auth + PostgreSQL; containerized **5-service** stack with Docker Compose for cloud deployment

### Bot Defender: Real-Time ML Traffic Classifier | [Live Demo](#) | *Python, FastAPI, Scikit-Learn, React*

- Engineered real-time anomaly detection pipeline with **sub-50ms inference latency**; trained **Random Forest** on **200K+ logs** from CIC-IDS-2017 for binary bot classification
- Architected **PostgreSQL (Supabase)** observability backend and **React** dashboard to visualize real-time confidence scores and threat probabilities

## EDUCATION

### Ganpat University

B.Tech Information Technology | GPA: 9.04 | **Departmental Rank 1**

Shree S.V. Shah VidhyaVihar / Nalanda

Higher Secondary (PCM) | Percentage: 69%

Mehsana, India

**August 2022 – June 2026**

Mehsana, India

**June 2021 – March 2022**

## CERTIFICATES

- Data Science Foundation – Infosys | [Link](#) (June '25)
- Crash Course on Python – Google | [Link](#) (Nov '24)
- Data Analysis with Python – Cognitive Class | [Link](#) (Dec '24)
- Data Analytics Job Simulation – Accenture | [Link](#) (Oct '24)